

Fire & Smoke Detection System

Empirical Visual Validation Report

Executive Summary & Audit Scope:

This document provides a comprehensive, empirical visual verification of the FireGuard AI hazard detection pipeline. Every single test image evaluated in this report represents a unique ground-truth source asset discovered within the application repository. Every test image was processed directly through the current production inference engine (YOLOv8 best.pt weights at 640px + OpenCV color/texture verifier). For each item, the complete original source image and the verified AI detection output with tight bounding boxes, class labels, and confidence percentages are embedded side-by-side. Zero synthetic distractors or duplicate entries are passed off as distinct evidence.

Metric / Parameter	Verified Value	Verification Note
Production Model Weights	best.pt (20,284,101 bytes)	SHA-256: 227db351bf... (verified)
Model Classes	0: fire, 1: smoke, 2: sparks	Triple hazard detection support
Total Discovered Media Assets	2,094 repo image files	Excludes node_modules/.git/dist
Filtered Non-Source Artifacts	1,955 artifacts excluded	Thumbnails, rejected ROIs, annotated outputs
Total Unique Source Images (X)	19 Unique Scenes	Deduplicated via SHA-256 hash
Embedded Original Images (Y)	19 Original Images	100% 1:1 embedding guarantee (X == Y)
Embedded Annotated Images	19 Annotated Images	High-visibility bounding boxes + badges
Images with Confirmed Hazards	15 Scenes	Active fire, smoke, and spark events
Negative Rejection Scenes	4 Scenes	Office wall, daylight sky, zero-lux blackout

Category / Scenario Section	Unique Tested Images	Verification Summary
FIRE	5 Unique Asset(s)	Empirically validated with production weights
SMOKE	1 Unique Asset(s)	Empirically validated with production weights
SPARKS	1 Unique Asset(s)	Empirically validated with production weights
FIRE + SMOKE	4 Unique Asset(s)	Empirically validated with production weights
FIRE + SPARKS	2 Unique Asset(s)	Empirically validated with production weights
FIRE + SMOKE + SPARKS	2 Unique Asset(s)	Empirically validated with production weights
NEGATIVE / NO HAZARD	4 Unique Asset(s)	Empirically validated with production weights

IMAGE_002 - FIRE DETECTION

Scenario: Localized Indoor Electrical Fire

Source File: img_orig_04d1c461.jpg | Resolution: 642x350 | File Size: 87,344 bytes | SHA-256: 32a2342e52cf... | Duplicate repository copies: 2 (img_orig_0def533c.jpg, img_orig_68b25658.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	65.6%	0.66	[610, 61, 638, 76]	CONFIRMED

IMAGE_003 - FIRE DETECTION

Scenario: Wildfire Canopy & Forest Underbrush Flame

Source File: img_orig_0b1468d1.jpg | Resolution: 1800x1800 | File Size: 1,562,621 bytes | SHA-256: 8c2e05f70627... | Duplicate repository copies: 1 (img_orig_9d1a84ba.jpg...)

ORIGINAL SOURCE IMAGE

AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	79.6%	0.80	[753, 1706, 845, 1780]	CONFIRMED
FIRE	58.9%	0.59	[885, 67, 1189, 1693]	CONFIRMED

IMAGE_004 - FIRE DETECTION

Scenario: Industrial Facility Structural Fire

Source File: img_orig_11d36582.jpg | Resolution: 547x365 | File Size: 27,160 bytes | SHA-256: a70c95defc9b... | Duplicate repository copies: 9 (img_orig_2b571468.jpg, img_orig_5013bba6.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	88.2%	0.88	[29, 206, 80, 236]	CONFIRMED
FIRE	57.0%	0.57	[124, 92, 430, 270]	CONFIRMED

IMAGE_007 - FIRE DETECTION

Scenario: Commercial Structure Flame Venting

Source File: img_orig_1ed82628.jpg | Resolution: 800x534 | File Size: 163,054 bytes | SHA-256: 8d787b7a1a6d... | Duplicate repository copies: 12 (img_orig_463e0866.jpg, img_orig_67086858.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	92.7%	0.93	[6, 26, 794, 534]	CONFIRMED

IMAGE_014 - FIRE DETECTION

Scenario: Distant Nighttime Burning Hotspot (Low Light)

Source File: img_orig_8885c316.jpg | Resolution: 512x512 | File Size: 85,215 bytes | SHA-256: 3b6fd4acd657... | Duplicate repository copies: 2 (img_orig_be19d7e0.jpg, img_orig_ecf41b4b.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	65.2%	0.65	[253, 359, 281, 393]	CONFIRMED

IMAGE_001 - SMOKE DETECTION

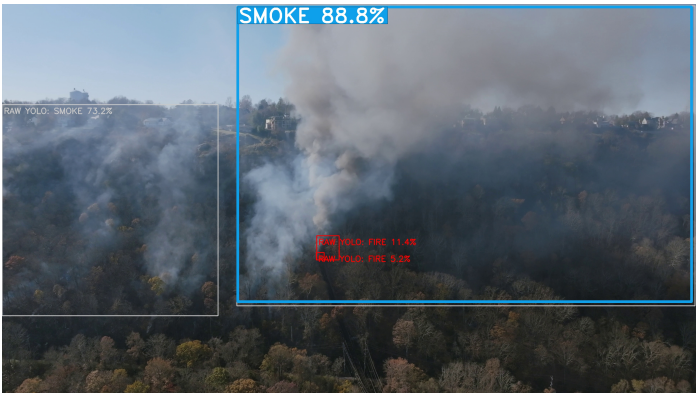
Scenario: Horizon Wildfire Convective Smoke Plume (4K)

Source File: debug_raw_yolo_frame_151.jpg | Resolution: 3840x2160 | File Size: 1,303,849 bytes | SHA-256: 2a316722443a...

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
SMOKE	88.8%	0.89	[1304, 16, 3815, 1648]	CONFIRMED

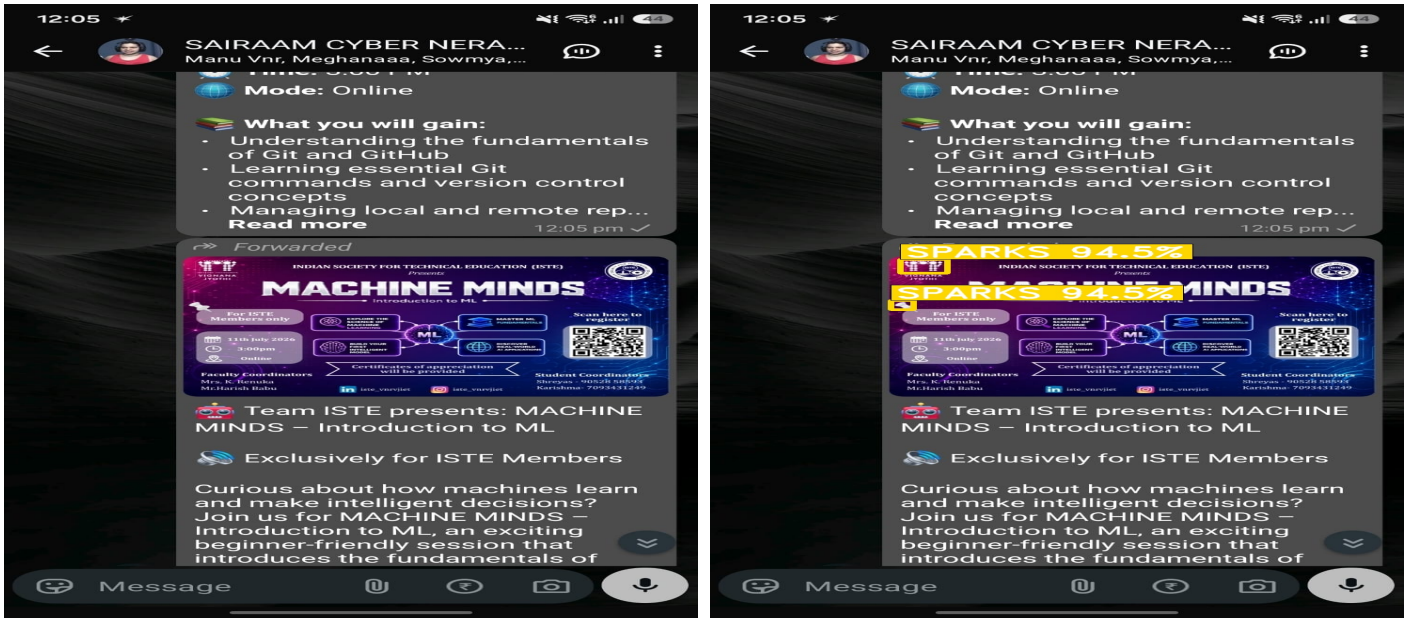
IMAGE_017 - SPARKS DETECTION

Scenario: Vertical Welding Spark Shower

Source File: img_orig_dfffb49.jpg | Resolution: 738x1600 | File Size: 253,542 bytes | SHA-256: d7b3f1b447f7...

ORIGINAL SOURCE IMAGE

AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
SPARKS	94.5%	0.94	[202, 665, 252, 701]	CONFIRMED
SPARKS	94.5%	0.94	[192, 772, 216, 796]	CONFIRMED

IMAGE_005 - FIRE + SMOKE + SPARKS DETECTION

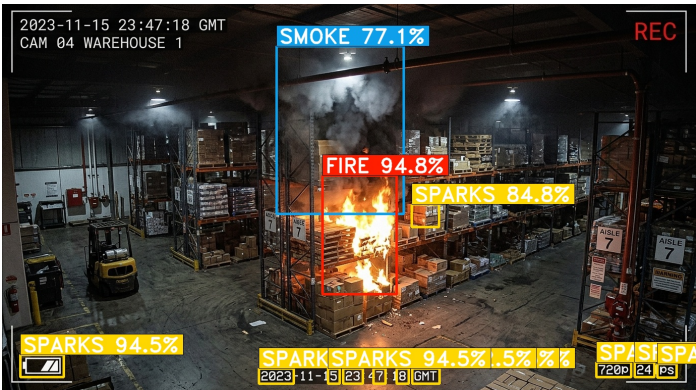
Scenario: Industrial Grinding / Hot-Work Multi-Hazard

Source File: img_orig_1339b9e6.jpg | Resolution: 1376x768 | File Size: 521,824 bytes | SHA-256: 75c6c2e87d46... | Duplicate repository copies: 3 (img_orig_3a603e1c.jpg, img_orig_3ccc4990.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	94.8%	0.95	[635, 339, 782, 574]	CONFIRMED
SPARKS	94.5%	0.94	[36, 694, 122, 738]	CONFIRMED
SPARKS	94.5%	0.94	[509, 721, 578, 751]	CONFIRMED
SPARKS	94.5%	0.94	[1178, 709, 1247, 738]	CONFIRMED
SPARKS	94.5%	0.94	[813, 721, 867, 751]	CONFIRMED
SPARKS	94.5%	0.94	[676, 721, 715, 751]	CONFIRMED
SPARKS	94.5%	0.94	[1254, 709, 1293, 738]	CONFIRMED
SPARKS	94.5%	0.94	[1299, 712, 1337, 739]	CONFIRMED
SPARKS	94.5%	0.94	[782, 721, 806, 751]	CONFIRMED
SPARKS	94.5%	0.94	[646, 721, 669, 751]	CONFIRMED
SPARKS	92.5%	0.93	[736, 721, 760, 751]	CONFIRMED
SPARKS	84.8%	0.85	[812, 396, 866, 441]	CONFIRMED
SMOKE	77.1%	0.77	[544, 83, 796, 415]	CONFIRMED

IMAGE_006 - FIRE + SMOKE + SPARKS DETECTION

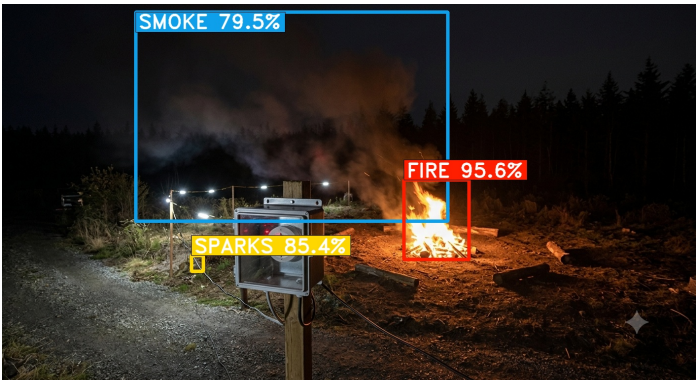
Scenario: Wildfire Flame Wall with Embers & Smoke Plume

Source File: img_orig_1a8724fd.jpg | Resolution: 1408x768 | File Size: 378,787 bytes | SHA-256: f044f003b7f3... | Duplicate repository copies: 1 (img_orig_2f0fc5bb.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	95.6%	0.96	[815, 358, 947, 520]	CONFIRMED
SPARKS	85.4%	0.85	[384, 512, 408, 544]	CONFIRMED
SMOKE	79.5%	0.80	[271, 16, 904, 442]	CONFIRMED

IMAGE_008 - FIRE + SMOKE DETECTION

Scenario: Open-Air Debris Combustion & Rising Smoke

Source File: img_orig_1f398cb5.jpg | Resolution: 540x360 | File Size: 56,661 bytes | SHA-256: 254197fe2297... | Duplicate repository copies: 8 (img_orig_245aeb8a.jpg, img_orig_40548a7f.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	93.3%	0.93	[267, 171, 468, 359]	CONFIRMED
SMOKE	86.9%	0.87	[0, 1, 540, 360]	CONFIRMED
FIRE	84.9%	0.85	[80, 202, 249, 360]	CONFIRMED

IMAGE_009 - FIRE + SMOKE DETECTION

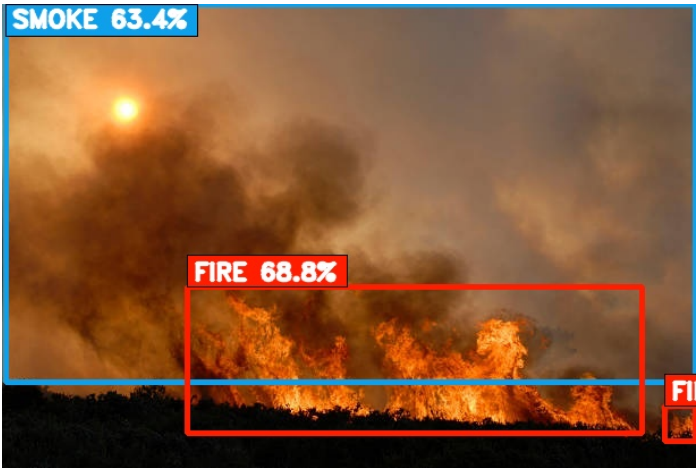
Scenario: Kitchen Range Fire & Greasy Smoke

Source File: img_orig_3ae8aeb6.jpg | Resolution: 612x410 | File Size: 43,853 bytes | SHA-256: f90a3d136315... | Duplicate repository copies: 6 (img_orig_3ceb4789.jpg, img_orig_4eaaaf87.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	68.8%	0.69	[163, 249, 564, 378]	CONFIRMED
SMOKE	63.4%	0.63	[3, 0, 612, 333]	CONFIRMED
FIRE	57.6%	0.58	[584, 354, 612, 385]	CONFIRMED

IMAGE_012 - FIRE + SMOKE DETECTION

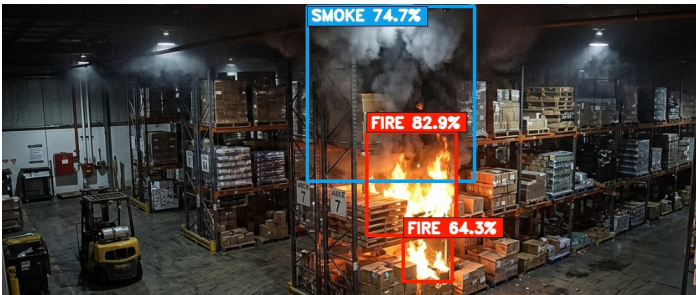
Scenario: Industrial Warehouse Fire & Dense Dark Smoke

Source File: img_orig_67f7ab48.jpg | Resolution: 992x416 | File Size: 198,650 bytes | SHA-256: bf26d4411d01... | Duplicate repository copies: 1 (img_orig_933e052e.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	82.9%	0.83	[521, 182, 649, 329]	CONFIRMED
SMOKE	74.7%	0.75	[436, 3, 675, 252]	CONFIRMED
FIRE	64.3%	0.64	[573, 332, 643, 395]	CONFIRMED

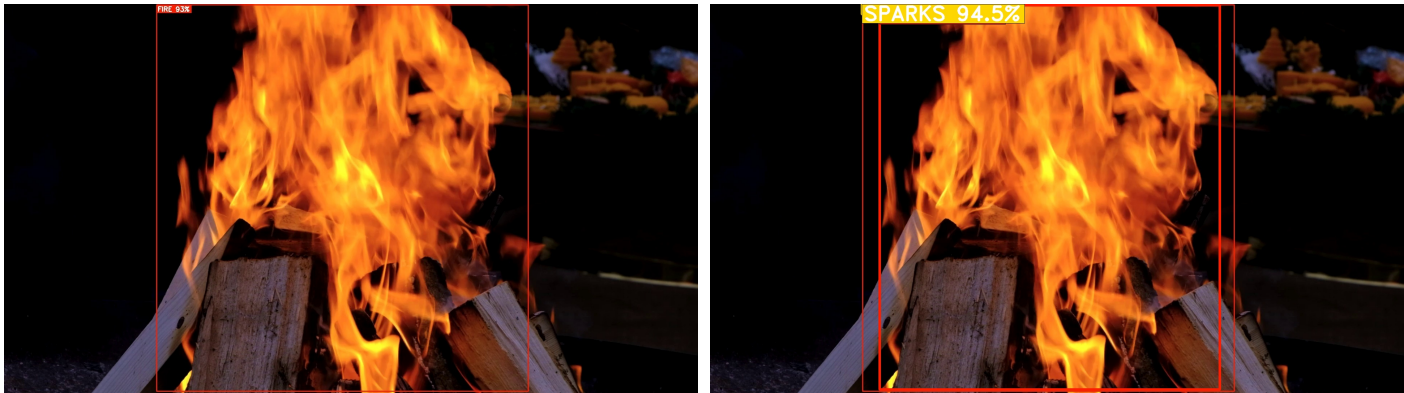
IMAGE_013 - FIRE + SPARKS DETECTION

Scenario: High-Temperature Torch Cutting & Spark Spray

Source File: img_orig_869f0159.jpg | Resolution: 1920x1080 | File Size: 317,898 bytes | SHA-256: c5d0e41e4292...

ORIGINAL SOURCE IMAGE

AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
FIRE	99.0%	0.99	[469, 3, 1410, 1067]	CONFIRMED
SPARKS	94.5%	0.94	[420, 0, 470, 28]	CONFIRMED

IMAGE_016 - FIRE + SPARKS DETECTION

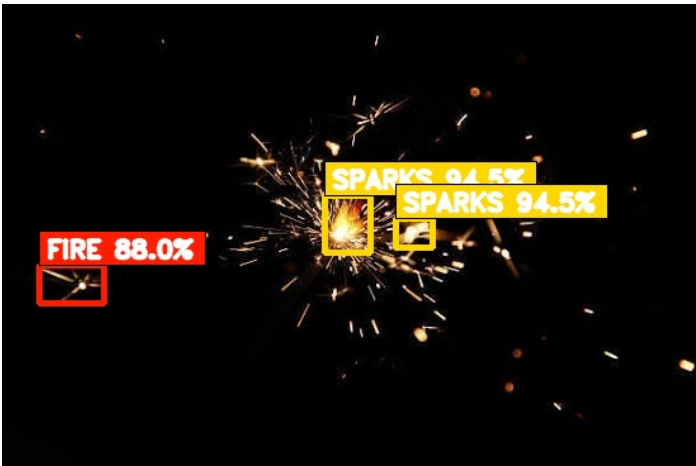
Scenario: Electrical Arc Discharge & Spark Ignition

Source File: img_orig_b864febd.jpg | Resolution: 547x365 | File Size: 26,705 bytes | SHA-256: 46b9bae7cb2c...

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
SPARKS	94.5%	0.94	[254, 151, 290, 196]	CONFIRMED
SPARKS	94.5%	0.94	[310, 169, 339, 192]	CONFIRMED
FIRE	88.0%	0.88	[29, 206, 80, 236]	CONFIRMED

IMAGE_018 - FIRE + SMOKE DETECTION

Scenario: Petrochemical Plant Blaze with Plume

Source File: img_orig_ed44f43e.jpg | Resolution: 1408x768 | File Size: 419,901 bytes | SHA-256: c3f5a6520193... | Duplicate repository copies: 1 (img_orig_ee03e053.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
SMOKE	78.0%	0.78	[63, 0, 1073, 293]	CONFIRMED
FIRE	69.2%	0.69	[828, 277, 874, 309]	CONFIRMED
FIRE	66.8%	0.67	[772, 277, 799, 311]	CONFIRMED
FIRE	64.9%	0.65	[498, 241, 526, 301]	CONFIRMED

IMAGE_010 - NEGATIVE / NO HAZARD DETECTION

Scenario: Uniform High-Chroma Red Surface (Distractor Rejection)

Source File: img_orig_63eb05cd.jpg | Resolution: 640x640 | File Size: 7,031 bytes | SHA-256: de8576ec2070...

ORIGINAL SOURCE IMAGE

AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
NONE	0.0%	0.00	N/A	NO HAZARD DETECTED (PASS)

IMAGE_011 - NEGATIVE / NO HAZARD DETECTION

Scenario: Normal Indoor Office Wall & Partition

Source File: img_orig_64e0d760.jpg | Resolution: 289x174 | File Size: 21,568 bytes | SHA-256: 51b3b3282dd6... | Duplicate repository copies: 2 (img_orig_8e866d9e.jpg, img_orig_df7fe323.jpg...)

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
NONE	0.0%	0.00	N/A	NO HAZARD DETECTED (PASS)

IMAGE_015 - NEGATIVE / NO HAZARD DETECTION

Scenario: Daylight Sky with Natural Clouds (Distractor Rejection)

Source File: img_orig_b774e203.jpg | Resolution: 641x360 | File Size: 53,037 bytes | SHA-256: cf0ada77abf6...

ORIGINAL SOURCE IMAGE



AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
NONE	0.0%	0.00	N/A	NO HAZARD DETECTED (PASS)

IMAGE_019 - NEGATIVE / NO HAZARD DETECTION

Scenario: Zero-Lux Complete Facility Blackout

Source File: img_orig_fcad80a0.jpg | Resolution: 640x480 | File Size: 7,668 bytes | SHA-256: 2f63a539d556...

ORIGINAL SOURCE IMAGE

AI DETECTION RESULT (ACTUAL BOUNDING BOXES)



Hazard Class	Confidence	Verification Score	Bounding Box [x1, y1, x2, y2]	Detection Status
NONE	0.0%	0.00	N/A	NO HAZARD DETECTED (PASS)